REST API Design for Online Store

Step 1: Conceptual Setup

Subtask 1.1: Resource Identification

The primary resources for the Online Store API are:

1. Products
    - Stores product information such as name, category, price, stock, and description.
2. Users
    - Stores customer account information such as username, email, and password.
3. Orders
    - Stores customer purchase orders and transaction details.

Additional optional resources:

- Reviews
- Categories
- Payments
- Cart

Subtask 1.2: Base URI & Versioning

To maintain backward compatibility and support future updates, the API will use versioning.

Base URI:

https://api.onlinestore.com/v1

Where:

- https://api.onlinestore.com = main API domain
- /v1 = API version 1

Step 2: Designing Read Operations (GET)

Subtask 2.1: Retrieve Collection

Endpoint to retrieve all products.

Endpoint:

GET /v1/products

Purpose:
Returns a list of all available products.

Subtask 2.2: Retrieve Single Resource

Endpoint to retrieve a single product using its unique ID.

Endpoint:

GET /v1/products/:id

Example:

GET /v1/products/101

Purpose:
Returns detailed information for product ID 101.

Subtask 2.3: Filtering, Sorting & Pagination

Endpoint with query parameters.

Endpoint:

GET /v1/products?category=electronics&sort=price_asc&limit=10

Purpose:

- category=electronics → filters products
- sort=price_asc → sorts by lowest price first
- limit=10 → only returns 10 products

Step 3: Designing Write Operations

Subtask 3.1: Create Resource (POST)

Endpoint to create a new order.

Endpoint:

POST /v1/orders

Example JSON Payload:

```
{
"userId": 1,
"products": [
{
"productId": 101,
"quantity": 2
}
],
"totalPrice": 1500
}
```

Purpose:
Creates a new customer order.

Subtask 3.2: Remove Resource (DELETE)

Endpoint to delete an order.

Endpoint:

DELETE /v1/orders/:id

Example:

DELETE /v1/orders/500

Purpose:
Deletes order ID 500.

Step 4: Hierarchy & Nested Resources

Subtask 4.1: Parent-Child Relationship

Endpoint to retrieve all orders belonging to a user.

Endpoint:

GET /v1/users/:userId/orders

Example:

GET /v1/users/1/orders

Purpose:
Returns all orders made by User 1.

Subtask 4.2: Reviews for a Product

Endpoint to retrieve reviews linked to a product.

Endpoint:

GET /v1/products/:productId/reviews

Example:

GET /v1/products/101/reviews

Purpose:
Returns all reviews for product ID 101.

Step 5: Final Review & Status Codes

Scenario	HTTP Status Code	Meaning
Successful GET request	200 OK	Request successful
Resource created successfully	201 Created	New resource added
Invalid request data	400 Bad Request	Client sent invalid data
Unauthorized access	401 Unauthorized	Authentication required
Forbidden access	403 Forbidden	Access denied
Resource not found	404 Not Found	Resource does not exist
Server error	500 Internal Server Error	Problem on server side
Successful deletion	204 No Content	Resource deleted successfully

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Conclusion

This REST API design follows standard RESTful conventions by:

- Using proper HTTP methods
- Applying versioning
- Organizing resources logically
- Supporting filtering and pagination
- Using appropriate HTTP status codes

The API structure is scalable, maintainable, and suitable for an Online Store application.